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R. F. Ellis, A. Case, R. Elton, J. Ghosh, H. Griem, A. Hassam, R. Lunsford, S. Messer, and C. Teodorescu



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Steady supersonically rotating plasmas in the Maryland Centrifugal Experiment^{a)}

R. F. Ellis,^{b)} A. Case, R. Elton, J. Ghosh, H. Griem, A. Hassam, R. Lunsford, S. Messer, and C. Teodorescu

University of Maryland, College Park, Maryland 20742-3511

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The Maryland Centrifugal Experiment MCX [R. F. Ellis, A. B. Hassam, and S. Messer, *Phys. Plasmas* **8**, 2057 (2000)] studies supersonic rotation and enhanced confinement produced by the application of an electric field perpendicular to an axial confining mirror magnetic field; radial shear in the rotation is predicted to stabilize magnetohydrodynamic (MHD) interchange modes. The MCX mirror field is 2.6 m in length, maximum mirror field 1.9 T, maximum midplane field 0.33 T; an inner coaxial core is driven by a 10 KV capacitor bank, producing the radial electric field which drives azimuthal rotation. MCX produces high density ($n > 10^{20} \text{ m}^{-3}$) fully ionized plasmas and has two operating modes. In the *O* (ordinary) mode the plasma rotates supersonically with azimuthal velocities in the range of 100 km/s for discharge times exceeding 8 ms. Ion temperatures are ~ 30 eV and momentum confinement times 100–200 μs . Sonic Mach numbers (u_ϕ/v_{ti}) in the range 1–2 and Alfvén Mach numbers (u_ϕ/v_A) ~ 0.3 have been achieved for *O* mode discharges which remain steady for many milliseconds, much longer than MHD instability time scales; plasma lifetime is limited by the capacitance of the capacitor bank. MCX also has an enhanced mode of operation [higher rotation (HR) mode] with higher rotation velocities (> 200 km/s), sonic Mach numbers greater than 3, Alfvén Mach numbers $> \sim 0.5$, and momentum confinement times of several hundred microseconds. HR mode occurs at higher *B* fields and lower discharge currents but is transient, transitioning to *O* mode after a few milliseconds. Both *O* and HR mode show spectroscopic evidence of radial velocity shear sufficient to satisfy the simplest criterion for MHD stability, but both modes also show significant fluctuations on magnetic probes.

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I. INTRODUCTION

Toroidal magnetic fields can contain the plasma pressures required for thermonuclear fusion, but for plasma equilibrium and stability the fields need to incorporate magnetic shear, which usually translates to complex or interlocking coil configurations. An alternative approach is to set the plasma in rapid rotation and utilize centrifugal forces to augment magnetic confinement, resulting in simpler coil configurations; this approach was pursued in earlier experiments.^{1–3} Moreover, recent observations of the suppression of plasma turbulence by flow shear^{4,5} have caused a revived interest in centrifugal confinement. Since rapid plasma rotation inevitably involves velocity shear, rotation could confine the plasma and stabilize it. Flow shear can be large enough to stabilize both large-scale magnetohydrodynamic (MHD) interchange modes (leading to simple coils) as well as small-scale turbulence (leading to superior plasma transport). In addition, viscous heating alone might bring the plasma to fusion temperatures, obviating auxiliary heating systems.

The schematic configuration is shown in Fig. 1. The *B* field has a simple mirror geometry with central solenoidal region. A coaxial metallic core is biased with respect to the

outer vessel. The radial electric field ionizes and forms the plasma and sets it into $\mathbf{E} \times \mathbf{B}$ azimuthal rotation. Radially outward centrifugal forces have a component parallel to *B* that contain the plasma to the solenoid region (the mirror loss cone is obviated if the rotation is supersonic). Plasma on field lines that end on conducting boundaries cannot rotate. Thus, only the plasma on field lines within the two shown as solid lines rotates, and plasma viscosity and drag adjust the electric field profile and the flow profile to result in sheared rotation. The simple magnetic mirror of Fig. 1 should be flute interchange unstable, on a time scale of the order of a rotation period if the rotation is rigid. For sheared rotation, however, the flute is expected to be stable if the flow is supersonic.⁶ The Maryland Centrifugal Experiment (MCX) (Ref. 7) is designed to access supersonic rotation and investigate centrifugal confinement and velocity shear stabilization of MHD instabilities.

II. EXPERIMENTAL SETUP

A. The MCX device

The MCX, Fig. 2, has the magnetic configuration of a long solenoid with axisymmetric mirror end fields. The length from mirror maximum to mirror maximum is 2.5 m with maximum field of 1.9 T; the independently controlled midplane field has a maximum of 0.33 T and the resulting

^{a)}Paper EI1 2, *Bull. Am. Phys. Soc.* **49**, 98 (2004).

^{b)}Invited speaker.

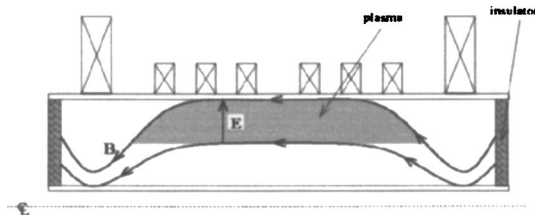


FIG. 1. Schematic of the centrifugal confinement scheme.

variable mirror ratio R_M is 2–20. The midplane plasma diameter is 0.54 m. The vessel is nonmagnetic stainless; except for the seals between the large sections all vacuum seals are copper gaskets; base pressure is 3×10^{-7} Torr. A stainless steel core (4.8 cm diameter), which is the high voltage electrode, runs along the axis. The core is insulated from the vessel by a custom designed high vacuum; high voltage feedthrough employing high purity alumina insulating tubes (5.0 cm inner diameter) which go over the core except for the central plasma region. Mounted on the alumina insulating tubes at symmetric axial points (which are variable) are two Pyrex disk insulators which terminate the plasma and prevent “shorting out” of the radial electric field; these insulating disks are located axially 0.37 m beyond the mirror maxima for the data in this paper. The insulating alumina tubes extend 0.31 m beyond the pyrex disks into the plasma region and the resulting “bare” core is 2.5 m long, corresponding to the mirror to mirror distance. Biasing of the stainless steel core relative to the outer wall drives $\mathbf{E} \times \mathbf{B}$ azimuthal rotation in the plasma region.

The high voltage is provided by a 10 kV ignitron-switched crowbarred capacitor bank, shown schematically in

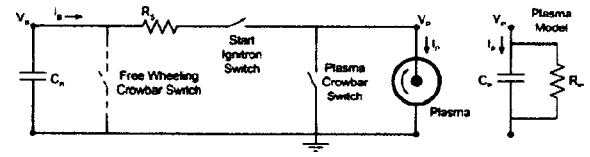
FIG. 3. MCX capacitor bank. $V_B=5\text{--}11$ kV, $C_B=1.3\text{--}7$ mF, $R_s=0.5\text{--}3$ Ω .

Fig. 3, along with the associated plasma model. The plasma crowbar allows us to terminate the discharge at any time by short circuiting both the capacitor bank and plasma to ground. Note that the circuit contains a series resistor R_s which limits the discharge current and plays a significant role in attaining higher rotation (HR) mode, as described in Sec. III B. For the experiments described here R_s is in the range 0.5–3 Ω with 0.5 Ω the standard for *O* mode operation.

B. Circuit measurements and model

The basic analysis of the plasma discharge employs data from the machine diagnostics (I_p and V_p versus time) and a zero dimension (0D) equivalent circuit model.² We model the rotating plasma as a capacitor C_p , in parallel with a plasma “leakage” resistance R_p (Fig. 3). Energy is stored in the dielectric medium as rotational kinetic energy. The resistor R_p represents net plasma momentum losses from viscous or friction forces; if the plasma capacitor were disconnected from the external voltage, the plasma rotational momentum would decay with a rate constant of $\tau_M = R_p C_p$. Based on these concepts, which are consistent with magnetized plasma dielectric theory and dissipative MHD, a set of model equations in zero dimension (0D) can be written down to describe the basic discharge. These are following:

$$\begin{aligned}
 (1/2)C_p V_p^2 &= (1/2)\rho u_\phi^2 \text{VOL} && \text{[energy in plasma capacitor = rotational kinetic energy]} \\
 C_p &= Q_p/V_p, \quad R_p = V_p/I_p && \text{[defines } C_p, R_p\text{]} \\
 E_r &= V_p/a_p, \quad u_\phi = E_r/B && \text{[} V_p \text{ and } a_p \text{ define average } E \text{ field across capacitor} \Rightarrow \text{inferred average } \mathbf{E} \times \mathbf{B} \text{ azimuthal speed, } u_\phi\text{]} \\
 \tau_M &= R_p C_p && \text{[plasma rotation damping given by the RC time constant of the leaky plasma capacitor]} \\
 1/\tau_{cx} &= n_0 \sigma_{cx} u_\phi && \text{[definition of } \tau_{cx}\text{; assumes supersonic flow]}
 \end{aligned}$$

Here u_ϕ is the azimuthal velocity, ρ is the plasma mass density, and VOL is the plasma volume. The charge stored in the plasma capacitor Q_p is directly measured by integrating the

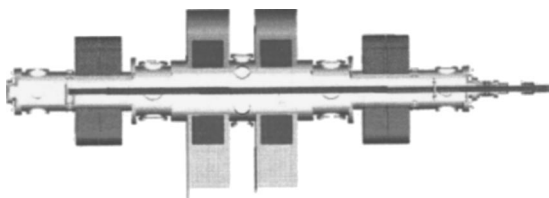


FIG. 2. Cutaway of the MCX device. For scale the mirror end coils are 1 m in diameter.

current reversal when the plasma is crowbarred (see Sec. III B). If the momentum decay is primarily due to resonance charge exchange of H^+ and H , represented by the cross section σ_{cx} , as concluded by Gerhard⁸ for parameters similar to MCX, we can estimate n_0 , the average neutral hydrogen density in the plasma by equating τ_{cx} with τ_M . The plasma radial extent at midplane a_p is determined by identifying the “last good flux surfaces” from the vacuum B field line mapping. The parameters in the model are meant to represent the highly ionized plasma annulus. For MCX I_p , V_p , Q_p are independently measured and u_ϕ is measured by Doppler spectroscopy for many discharges. *O-D* model determinations of plasma density were confirmed by comparison with

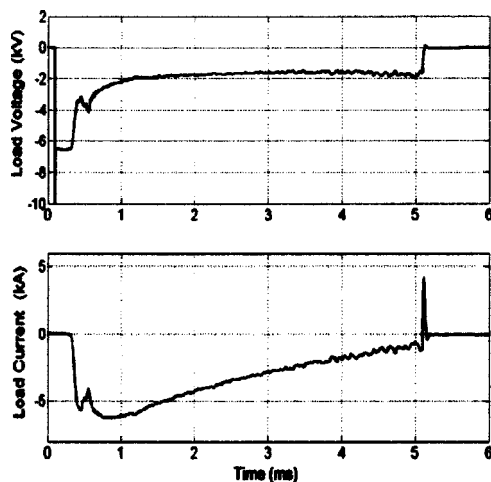


FIG. 4. Typical *O* mode discharge. $C_B=3.5$ mf, $V_B=6.5$ kV, midplane $B=0.2$ T, $R_M=9$, $R_S=0.5$ Ω .

direct measurements of density employing a HeNe interferometer.⁹ (Section II C). Some limitations of the 0D model are discussed following Fig. 7 and we emphasize it is a tool as to determine approximate quantities and scalings.

We note that the basic model behavior of the actual MCX circuit was tested by discharging the capacitor bank into a variety of dummy loads with parameters similar to the R_p and C_p of the MCX plasma. In all cases the circuit behavior was consistent with our analysis, in particular, in verifying that the charge in the current reversal pulse (Fig. 4) is an accurate measurement of the stored charge in the capacitor at crowbar and could be used to calculate the stored energy. These measurements also determine the inductance in the external portion of the MCX circuit which, as designed, is very small (~ 1 μ H).

C. Other diagnostics

Visible spectroscopy is employed to directly measure rotation velocity (Doppler shift) and ion temperature (Doppler broadening) for H_{α} and impurity lines (C^{1+} – C^{3+} , N, Si) in the plasma.⁹

Magnetic probes located near midplane and near mirror maximum (located on approximately the same B line, at the plasma edge) are used to measure magnetic fluctuations. The probes can independently measure three orthogonal components of the fluctuating B fields.

A visible ($\lambda=633$ nm) HeNe interferometer was borrowed from the surface soft-x-ray experiment at Swarthmore College and electron density was measured for a wide range of discharge parameters. For all cases these were in good agreement with the volume averaged plasma density calculated from the circuit model described in Sec. II B.⁹ In applying the model we assumed the radial velocity profile was parabolic (see Fig. 7) and ignored any changes due to plasma β . More details can be found in Ref. 9. Recent Stark broadening measurements of Balmer series spectral lines¹⁰ confirm the density measurements.

TABLE I. MCX operating range.

Core polarity	Pressure (m Torr)	Voltage (kV)
Negative	0.5–20	3–10
Positive	3–20	5–10

III. EXPERIMENTAL RESULTS

MCX routinely generates high density, significant β , fully ionized plasmas rotating supersonically. The plasma generation is robust and reproducible, and the plasma lifetimes vastly exceed MHD instability time scales, being limited only by the available power source (capacitor bank). Recently, in addition to our standard or *ordinary* mode of operation, referred to as “*O* mode,” we have discovered a new *high rotation*, higher performance mode, referred to as “HR mode”. We begin by noting that for many discharges the ion rotation velocity is a quantity directly measured by Doppler spectroscopy, as is the ion temperature. Many other quantities are computed from these and from the *O-D* model for which detailed profile information is not in hand. The *O-D* model calculation of volume averaged plasma density was confirmed by the aforementioned interferometer and visible spectroscopy measurements.

A. Plasma creation and operating parameter space

Plasma is created by breakdown and ionization from the external applied voltage. The initial electric field has a component parallel to the magnetic field, accelerating electrons along the field, leading to breakdown.

We have been able to generate rotating plasmas with the center core both positive and negative, for a range of static fill pressures of hydrogen and various voltages, as shown in Table I. Lower pressures generally require higher voltages for breakdown and the upper pressure limit is simply the highest pressure we investigated. Also note that the ranges are quite wide, especially for a negative core. All the data presented in this paper are for negative polarity.

B. Standard *O* mode discharge

Figure 4 displays the plasma voltage V_p and current I_p for a typical MCX *O* mode discharge. After a holdoff and formation time of ~ 1 ms the plasma voltage settles at about 2 kV and holds steady, while the current slowly decreases to about 1 kA. At $t \approx 5$ ms the discharge is terminated by the plasma crowbar, which is followed by a strong reversal pulse in the plasma current, representing the stored energy in the rotating plasma. Time integration of the reversal current yields the equivalent charge Q_p stored in the rotating plasma. We note that I_p can be held more constant by increasing the bank capacitance, and from our experiments we conclude that the MCX *O* mode discharge may remain in a steady state, as long as the plasma voltage and current can be maintained approximately constant.

Summarizing the *O* mode results briefly, for a wide range of parameters the data and the 0D model, in combination with Doppler spectroscopy and supported by limited

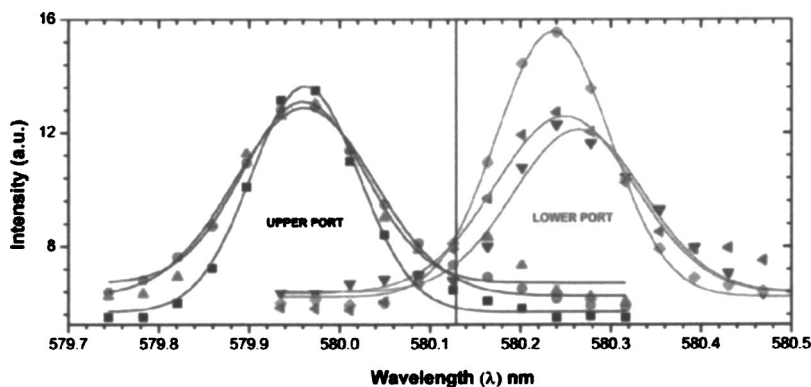


FIG. 5. C^{3+} spectra measured for two chords through the plasma. Three identical discharges are shown for each chord.

HeNe interferometer and spectroscopic density measurements, show that O mode plasmas are high density ($n > 10^{20} \text{ m}^{-3}$), rotating supersonically with peak $u_\phi \sim 100 \text{ km/s}$; ion temperatures are typically 20–40 eV. Doppler measurements of u_ϕ for various chords confirm the $\mathbf{E} \times \mathbf{B}$ nature of the rotation. Sonic Mach numbers [$M_S = u_\phi / (T/M_p)^{1/2}$] are in the range 1–2, with Alfvén Mach numbers ($M_A = u_\phi / v_A$) typically ~ 0.3 , with peak thermal β 's of $\sim 20\%$. Assuming that all the rotation damping is from friction due to charge exchange with neutrals, we can infer an upper bound on the neutral density in the plasma, which tells us that the plasma is very highly ionized ($n_0/n_i < 0.01$). The corresponding O -D model momentum confinement time, $\tau_M = R_p C_p$, is in the range 100–200 μs . Note we do not conclude from this neutral density argument that fluctuations or turbulence are not important in determining the momentum confinement time. Making this determination is one of our future goals.

We now turn to selected data demonstrating many of these points and begin with Doppler spectroscopy.¹¹ The Doppler spectroscopy measurements mentioned above were made of impurity lines (C^{1+} – C^{3+} , N, Si) and H_α viewed along the chord 16.5 cm off-axis above the vessel centerline and for identical conditions measurements were made along a chord diametrically opposite to the first view, below the centerline. Figure 5 shows typical spectra for the C^{3+} line at a wavelength of 580.13 nm from different, reproducible, MCX discharges (mirror ratio=9, applied voltage=–7 kV, fill pressure=5 mTorr, and midplane magnetic field=0.1 T). Fitted Gaussian line shapes to the experimental data are also shown which yield the line shift (velocity) and width (ion temperature). The spectra were collected at 550 μs into the discharge (integration time=100 μs). There are clearly large Doppler shifts (the solid vertical line is the unshifted line), and the top-bottom shifts are opposite to each other as expected for $\mathbf{E} \times \mathbf{B}$ rotation. Observations were also made with the sign of the magnetic field reversed and the results were consistent with $\mathbf{E} \times \mathbf{B}$. Finally, the null measurement was made viewing radially, directly at the core. In this case the Doppler shift yielded a velocity of less than the experimental error bar of 10 km/s. The Doppler shifts show an average rotation speed ranging from 60 km/s to over 100 km/s for the O mode. Doppler broadening gives an ion temperature in the range of 20–40 eV, assuming the impurity ion temperature is similar

to that of the bulk plasma. These velocities and temperatures correspond to an ion-thermal Mach number, $M_S = (M_p u_\phi^2 / T)^{1/2}$, ranging from 1 to 2.

The fact that we have a radial rotation profile with shear can be inferred from Fig. 6, which shows measured Doppler rotation velocities versus ionization state for a number of impurity lines for both O and HR mode plasmas. Because higher ionization states can penetrate further into the highly ionized MCX plasma, these results are direct evidence of a reasonably smooth velocity profile with significant shear. Note also that Fig. 6 shows that the HR mode is rotating more than twice as fast as the O mode.

Confirmation of the O -D model calculation of u_ϕ is provided in Fig. 7, where we show the results of a comparison of spectroscopically determined rotation velocities with $V_p / a_p B$ for a scan of the MCX magnetic field magnitude at a fixed mirror ratio $R_M = 9$ for O mode. Only the C^{3+} results are shown as C^{3+} is most representative of the rotation at plasma center (see Fig. 6). We note first that above about 0.9 kG the plasma rotation velocity is approximately constant with increasing B , consistent with the fact that as B increases the measured V_p increases. Second, two calculations of $V_p / a_p B$ are shown, both of which assume the plasma radially fills the available flux surfaces. However, the lower set assumes the velocity is uniform in radius while the upper set assumes a parabolic profile and the peak value is plotted. Note that the spectral values are bounded by the two and that the parabolic case gives a much better fit at higher B where most MCX

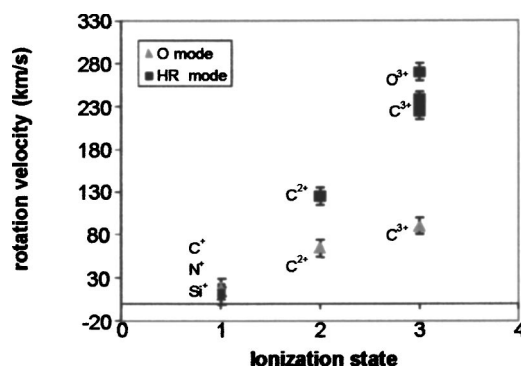


FIG. 6. Doppler ion rotation velocity vs impurity ionization state for both O and HR mode plasmas.

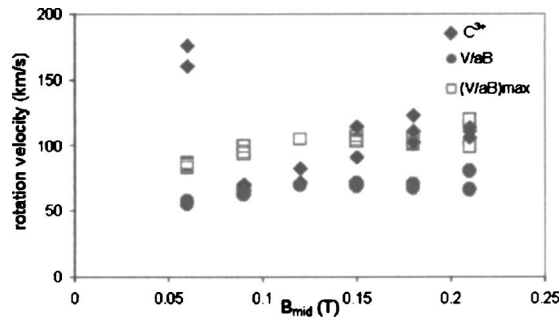


FIG. 7. Ion rotation velocity vs midplane B (O mode, $R_M=9$, $V_B=7$ kV). C^{3+} from spectroscopic measurement, V/aB from 0D model with flat radial profile, $(V/aB)_{\max}$ from 0D model with peak value of a parabolic profile.

data are taken. These data, and many more discharges for which we have spectroscopy, confirm that $V_p/a_p B$ is a good measure of the rotation speed for reasonable choices of a_p and rotation profile. It is appropriate here to discuss some limitations to the 0D model and the determination of a_p which is the radial extent at plasma midplane. First, a_p is estimated from vacuum magnetic field line mapping. The outer radial plasma boundary is fixed by the field line which intersects the vacuum vessel at the outer wall; the inner boundary by the field line which grazes the core in the region of the magnetic mirror. Because our plasma has significant rotational β there may be field line distortion at the highest β (bowing). However, the outer boundary is more or less fixed by the vessel. For the highest rotation speeds the inner boundary may be displaced a small amount but a radial β profile almost certainly has low β at the inner boundary and the issue really is profile shape. In any case this β effect has

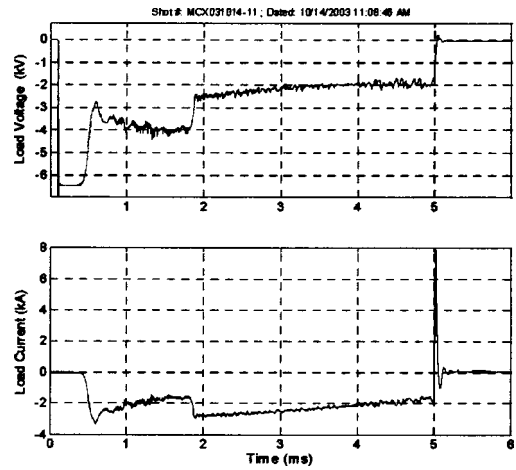


FIG. 9. Discharge with HR and O mode (midplane $B=0.2$ T, $R_M=9$, $R_s=1.5 \Omega$).

not been included here. Second, a_p is determined from the type of data shown in Fig. 7 comparing spectroscopic measurements with the model. This determination is of course affected by our choice of rotational profile and Fig. 7 gives an estimate of this effect. Field line bowing may also play a role here at the highest β because the vacuum value of B is used in this calculation. We also note that the two large spectroscopically measured velocities at lowest B are for discharges with very little stored energy and were not investigated further; MCX is generally not operated at these low fields.

Considering the measured Mach numbers, in Fig. 8 are shown histograms of the distribution of measured sonic and Alfvén Mach numbers for a large number of O mode and HR mode discharges. Azimuthal velocities are center values for an assumed radial parabolic profile to agree with Doppler measurements. Note both O and HR mode are supersonic with a clear separation of the two modes. Also note HR mode achieves Alfvén Mach numbers close to 1.

Turning to the magnetic fluctuations, in O mode these have $\delta B/B$ of about 0.1% and the frequency spectrum has significant activity at the edge rotation frequency and does not exclude Alfvén eigenmodes with parallel wavelengths comparable to the machine length.

C. Higher rotation (HR Mode) discharge

We turn now to the new mode of operation which we call HR mode. The HR mode appears when discharge current is limited, by increasing the resistor R_s in series with the plasma (typically $0.5\text{--}4 \Omega$) and/or increasing the strength of the magnetic field, both causing an increase in effective plasma resistance. HR mode is characterized by higher rotation velocities (200–300 km/s), longer momentum confinement times (200–400 μs), higher resistance, and larger magnetic fluctuations. All HR modes are transient, appearing just after formation and transitioning to the O mode after some milliseconds or less. However, quasisteady HR discharges for durations of 3–4 ms are common.

Figure 9 shows a typical discharge exhibiting HR mode. This discharge was run at our standard high B but with a

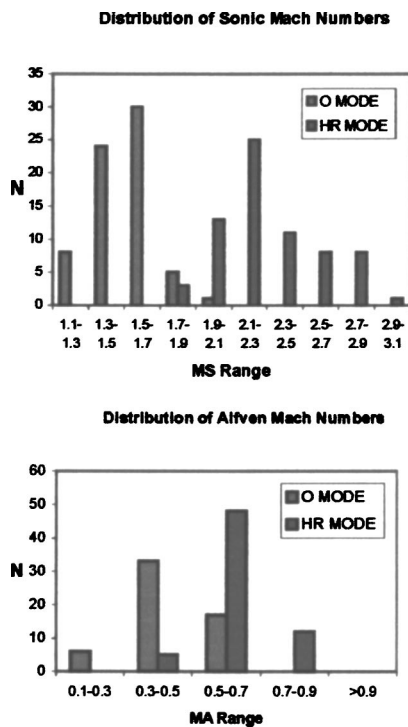


FIG. 8. Distribution of Mach numbers for a set of O and HR mode discharges.

TABLE II. Parameters bracketing HR to *O* mode transition.

Discharge mode	Rotational velocity (km/s)	Ion density (10^{20} m^{-3})	Stored energy (kJ)	Momentum confinement time (μs)
HR ($t=1.6 \text{ ms}$)	248	1.7	1.08	260
<i>O</i> ($t=2.0 \text{ ms}$)	98	2.8	0.31	80

series resistor of 1.5Ω (rather than 0.5Ω) which limits the plasma current. Note that for times up to about 2 ms the voltage across the plasma is almost 4 kV but the current is less than 2 kA, indicating a plasma resistance exceeding 2Ω . At $t \sim 2 \text{ ms}$ the mode transitions to the usual *O* mode described earlier. A cursory look at HR mode shows it is characterized by higher rotation velocities (higher V_p) and higher resistance (higher V_p and lower I_p).

In general, the HR mode appears when the discharge current is limited, by increasing the resistor R_s in series with the plasma (typically $0.5\text{--}4 \Omega$) and/or increasing the strength of the magnetic field, both causing an increase in effective plasma resistance. In general HR mode occurs for midplane B of 0.2 T or greater and R_s from about $1\text{--}3 \Omega$. Increasing R_s beyond about 3Ω leads to a weak discharge. The maximum possible midplane B is presently 0.33 T. All the data we have taken support the hypothesis that limiting discharge current is the key to HR mode operation.

HR discharges show a factor of order 2–3 increase in rotation velocity, stored energy, and confinement time compared to the *O* mode. This is confirmed by firing the plasma crowbar just before transition (HR mode) in one discharge and just after transition (*O* mode) in another identical discharge and by calculating the plasma capacitance C_p , the energy stored in the equivalent plasma capacitor $0.5Q_pV_p$, and the momentum decay time R_pC_p . The results are shown in Table II for six such discharges; note that these results do not employ any profile information. The improved performance in HR mode is striking.

Referring back to Fig. 8, we note that it shows the distribution of Mach numbers for a large number of HR mode discharges along with the *O* mode results. We see again a clear separation of the two modes and note importantly that the HR mode achieves sonic Mach numbers exceeding 2 and Alfvén Mach numbers greater than 0.5. Confirmation of the high rotation velocity in HR mode was obtained with measurement of the Doppler shifted C^{3+} emission for the HR and *O* portions of a discharge. A typical velocity from the Doppler shift gives 200–300 km/s in HR mode. T_i was measured from Doppler broadening and is about 20% higher in HR mode. Finally, HR plasmas are also high density ($>10^{20} \text{ m}^{-3}$) and can be characterized as very highly ionized. Also of importance, magnetic probe activity shows edge magnetic fluctuations in the HR mode which are more than three times larger than in the *O* mode, with a frequency spectrum shifted to higher frequencies consistent with the higher rotation velocities. This could be related to the higher

TABLE III. Characteristic mcx time scales. $B=0.2 \text{ T}$, $a_p=0.2 \text{ m}$, $L_p=1.6 \text{ m}$, $n=10^{20} \text{ m}^{-3}$, $T_e=T_i=30 \text{ eV}$, $u_\phi=100 \text{ km/s}$.

Axial Alfvén time $\sim L_p/V_A$	5 μs
Period of rotation $\sim (2\pi R/u_\phi)$	10 μs
Interchange growth time $\sim [(a_p L_p)/(T/M_p)]^{1/2}$	10 μs
Axial electron heat conduction time $\sim (L_p/\lambda)^2 \tau_e$	15 μs
Axial sonic time $\sim L_p/(T/M_p)^{1/2}$	30 μs
Electron-ion heat exchange time $\sim (M_p/m_e) \tau_e$	80 μs
Classical viscous damping time $\sim (a_p/\rho)^2 \tau_{ii}$	8000 μs

Alfvén Mach number, which brings MCX closer to an MHD stability boundary. It is interesting that larger confinement time correlates with higher velocity shear but also larger fluctuations. It may be that more shear leads to better confinement through some mechanism other than stabilization of MHD modes.

IV. TIME SCALES AND STABILITY

The relatively quiescent and “steady state” current and voltage traces of Fig. 4 and the *O* mode portion of Fig. 9 show steady rotation over long periods. Characteristic plasma times for typical MCX parameters are shown in Table III.

If MCX were unstable to MHD interchanges, the instability growth time scale could be as short as a rotation period and not longer than an interchange growth time, both of the order of 10 μs , whereas the MCX I-V traces hold steady up to 8 ms, being limited only by the crowbar or by the finite charge in the external capacitor bank. In addition, if there were large unstable flute modes in the discharge, momentum and heat would be lost from the discharge on time scales as short as the flute rotation period, but the observed damping time of the rotation R_pC_p is measured to be 100 μs for the *O* mode. For the HR mode the momentum damping time is greater than 200 μs and the difference is even larger.

Note that the classical damping time of the rotation from ion-ion collisions is $\approx 8 \text{ ms}$ —there is a clear separation of ideal MHD instability times and dissipative time scales. The plasma sustenance over long time scales and the clear separation of dissipative and ideal time scales is a strong indication that the plasma is not violently unstable to MHD flute modes. At present there is no direct evidence that stability results from rotational velocity shear but both *O* and HR modes satisfy the simplest condition for shear stabilization that the radial shear exceed the instability growth rate ($du_\phi/dr > \gamma_{\text{MHD}}$) if we take $d/dr \sim 1/a$. The variation of rotation velocity with ionization state, as measured by Doppler spectroscopy, supports the proposition that such a radial variation exists. We also note that if instability induced turbulence losses are small, then the observed momentum confinement time might be mostly attributable the ion-neutral charge exchange collisions, and as noted in Sec. II B we can estimate an upper bound on the neutral hydrogen density in the main body of the rotating plasma. This calculation results in an ionized fraction greater than 99%. However, a more complete model of neutral transport is needed to establish

this with any certainty. This is not to imply that we conclude fluctuation driven losses caused by MHD activity or micro-turbulence are small only that this assumption gives us an upper bound on neutral density.

V. CONCLUSIONS

The Maryland Centrifugal Experiment has demonstrated the generation of supersonically rotating plasmas in a mirror geometry. The plasmas are high density and highly ionized. Two discharge types exist—the standard or *O* (ordinary) mode and the higher performance HR (high rotation) mode. *O* mode discharges are steady (stationary) for multi-millisecond durations with the duration being limited by capacitor bank stored energy. HR mode discharges are stationary for a duration of a few milliseconds but then transition to *O* mode. The momentum confinement times in both modes greatly exceed the MHD instability growth time but are much less than predicted by classical diffusion. These confinement times and discharge durations show the plasmas are not unstable to large amplitude MHD flute modes. While spectroscopic evidence indicates the rotational velocity has substantial shear we cannot yet conclude that any stabilization is a result of this shear. In addition, other instabilities, turbulence, or neutral drag cannot be ruled out at this time as limiting the momentum confinement time.

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